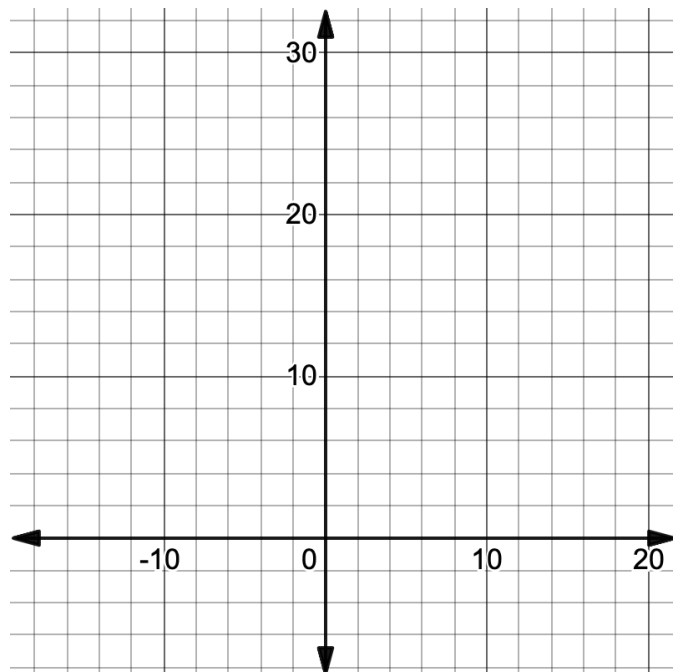




11.3.6 Wrap-Up: Ticket Out the Door

1. Graph the function $y = 9\left(\frac{5}{3}\right)^x$.



Unit 11, Lesson 4: Graphing Exponential Functions Given a Written Description and Equation



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



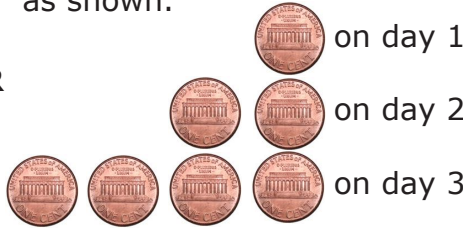
Learning Targets

- ☐ I can graph an exponential function given a written description and equation.
- ☐ I can determine the key features of an exponential function.
- ☐ I can classify a function as representing exponential growth or decay.



11.4.1 Warm-Up: Would You Rather?

- Imagine you have the chance to pick from one of the two options shown.

Option A	Option B
Receive a one-time payment of \$1 million dollars.	Receive an increasing number of pennies each day as shown.
	OR and so on for 1 month

- Would you rather choose Option A or Option B?
- Using mathematics, show which option pays out more money.

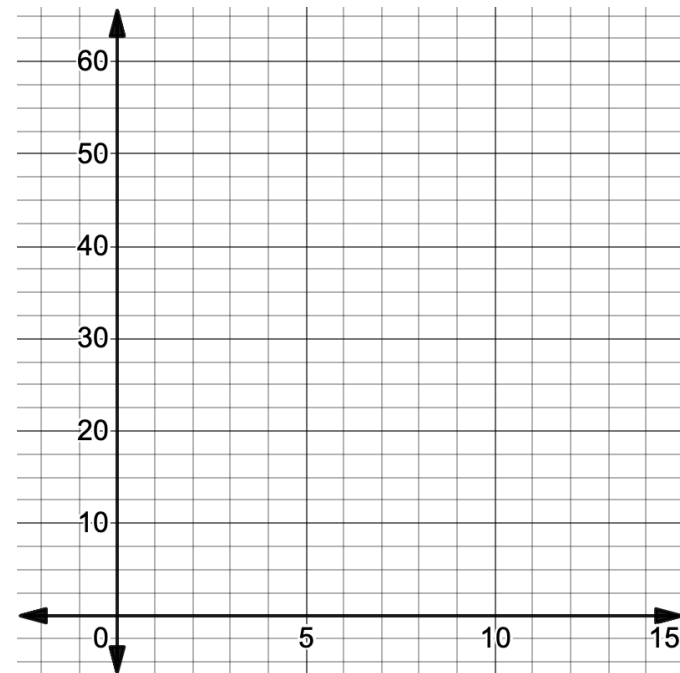


11.4.2 Guided Instruction: Graphing Exponential Functions From a Written Description and Equation

The half-life of a drug is an estimated period of time that it takes for an amount of that drug in the body to be reduced by exactly half. The amount, A , in milligrams, of a drug in the body for h hours after taking a pill is given by

$$A(h) = 50 \left(\frac{1}{2} \right)^h.$$

- Graph the function that models the situation.



- Describe how to determine whether the situation models exponential growth or decay.

3. What is the initial dose given?

4. Describe how to identify the initial dose from the function and from the graph in the table.

From the Function	From the Graph

5. What percentage of the drug leaves the body each hour?

6. How is this percentage represented in the equation?

7. Complete the table to identify other key features of the function based on the context. In the second row, explain how each key feature relates to the context.

	Domain	Range	As x -values increase, y -values ...
Key Feature			
Reason for Constraint from Context			





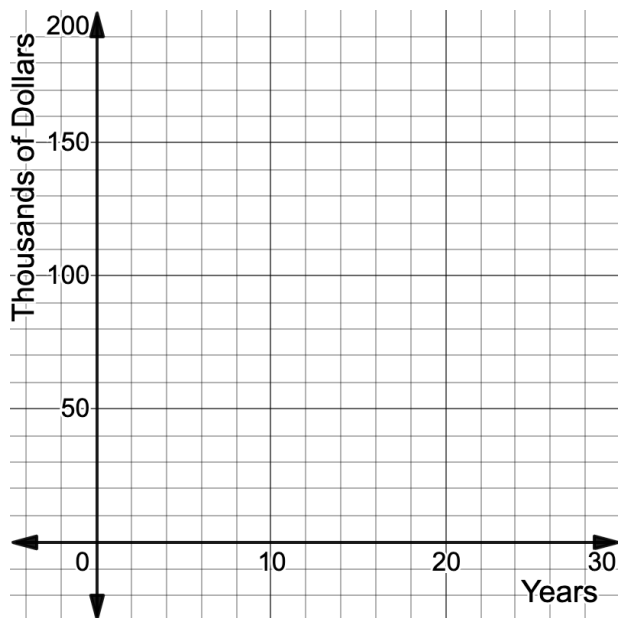
11.4.3 Collaboration: Graphing Exponential Functions

Liam recently graduated from the University of South Florida and accepted a job as a software engineer. The starting salary is \$50,000 and has a guaranteed annual pay increase of 3% each year.

The equation $y = 50,000(1 + 0.03)^x$ models the increase in salary over time, where x is the number of years since he started at the company and y is his salary.

1. Describe how to determine if this situation models exponential growth or decay.

2. Graph the equation $y = 50,000(1 + 0.03)^x$.



3. What does the y -intercept represent in this situation?

4. Harper said that the domain is $0 \leq x \leq \infty$. Discuss with your partner if Harper is correct. Write a summary of your discussion.

5. Describe why the range depends on the number of years the person works for the company.

6. Complete the statements.

If Liam works for the company for 30 years, the domain is _____ $\leq x \leq$ _____, with a range of _____ $\leq y \leq$ _____.

7. Two students were asked to find the constant percent rate of change for this situation. Their answers are shown.

Owen	Fiona
The constant percent rate of change for the situation is 3% because Liam receives a pay increase of 3% every year.	The constant percent rate of change is 103% because each year Liam receives a 3% pay increase in addition to 100% of the salary he earned the year before.

- Discuss with your partner who identified the correct constant percent rate.
- Explain the mistake made by the student who was incorrect.



11.4.4 Activity: Gallery Walk

- Work with a partner to graph the situation below that your teacher has chosen for you.

Situation A	Situation B
The annual rate of inflation for goods and services is modeled by $C(t) = P(1.024)^t$, where C is the approximate cost of goods or services, t is the time in years, and P is the present cost. If the present cost of Nike LeBron basketball shoes is \$149.99, graph the function that will model the cost of shoes over time.	A new 2021 Ford Ranger XLT costs \$31,045 with a depreciation rate of 8% per year. This situation is modeled by the function $V(x) = 31,045(1 - 0.08)^x$, where x is the number of years since 2021 and V is the value of the car.
Situation C	Situation D
One hundred bacteria cells are put into a petri dish, and the culture doubles in size every hour. This is modeled by $P(h) = 100(2)^h$, where P is the number of bacteria in the population and h is the number hours that have passed.	A new computer that was recently purchased depreciates over time by the given function, $C(x) = \$1,200\left(\frac{1}{4}\right)^x$, where x is the number of years since the computer was purchased and C is the current value of the computer.

2. Once you and your partner graph the function, determine the key features and interpret how each relates to the context.

	Key Feature	Interpretations
Domain		
Range		
y-intercept		
Constant Percent Rate of Change		
End Behavior		



11.4.5 Wrap-Up: Growth Mindset

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always